

Research

Artificial intelligence-driven integration of single-cell RNA sequencing and transcriptome analysis to decipher APOE's role in gastric cancer prognosis and therapy

Ni Li¹ · Jinquan Lin^{2,3} · Chunyu Wang³ · Xiao Ran³ · Zhi Zhao²

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Abstract

Background Gastric cancer (GC) remains a global health challenge due to its high mortality rate and the lack of specific diagnostic methods. Tumor heterogeneity significantly hinders effective treatment, necessitating advanced techniques to dissect its complexity. Artificial intelligence (AI) offers transformative potential in uncovering intricate cellular dynamics and gene regulatory networks. This study leverages single-cell RNA sequencing (scRNA-seq) combined with transcriptome analysis, enhanced by AI-driven analytics, to explore the tumor microenvironment and identify novel prognostic markers and therapeutic targets in GC.

Methods scRNA-seq and transcriptome datasets of GC patients were analyzed using AI-enhanced methodologies to unravel tumor heterogeneity and microenvironmental dynamics. Macrophage subsets were identified as critical components within the GC microenvironment. High-variance gene screening in these subsets pinpointed apolipoprotein E (APOE) as a hub gene. Experimental validation of APOE expression in GC samples and functional studies in GC cell lines were conducted.

Results Bioinformatics and AI-enabled analyses confirmed the immunosuppressive role of APOE in GC. An immune-related survival model was developed to predict immunotherapy responses and patient prognoses. Mechanistically, APOE was found to induce immunosuppression through M2 macrophage polarization, promoting tumor progression and leading to poorer outcomes in GC patients.

Conclusion This study highlights the potential of AI-driven approaches in elucidating the role of APOE in GC progression. APOE's regulatory effects on M2 macrophages underscore its value as a prognostic marker and therapeutic target, paving the way for precision medicine in GC management.

Keywords Artificial intelligence · APOE · Gastric cancer · Tumor immune microenvironment

Ni Li, Jinquan Lin and Chunyu Wang contributed equally.

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✉ Zhi Zhao, xykz521@163.com | ¹Health Management Center, People's Hospital of Guilin, Guilin 541002, Guangxi, China. ²Department of Gastrointestinal and Hernia Surgery, People's Hospital of Guilin, Guilin 541002, Guangxi, China. ³Guilin Medical University, Guilin 541199, Guangxi, China.



1 Introduction

Gastric cancer (GC) is one of the most common malignancies worldwide. Due to the lack of specific symptoms in the early stage, GC is often in an advanced stage at the time of initial diagnosis, more than 700,000 people die of GC worldwide each year [1]. At present, the diagnosis and treatment of early GC mainly rely on esophagogastroduodenoscopy [2], and advanced GC are mainly treated with surgery, chemotherapy, radiotherapy, and immunotherapy [3, 4]. Although the median survival time of GC patients have increased due to the development of various treatment approaches, the overall survival (OS) rate is still very low [5].

During the growth process of tumor, daughter cells exhibit molecular biological or genetic differences from the parent cells after division and proliferation that can affect the growth rate, invasiveness, drug sensitivity, and prognosis of the tumor, which is called tumor heterogeneity and one of the main characteristics of malignant tumors [6]. Because traditional sequencing analysis cells in bulk, what we actually obtained is the mean value of the signals in many cells, the information on cell heterogeneity is lost. Single-cell RNA sequencing (scRNA-seq) is a technology used for sequencing and analyzing the genome, transcriptome, and epigenome at the level of individual cell. Which can provide a statistical approach for the characterization of different cell subsets in tumors and is an important tool for the study of cell heterogeneity [7]. Important information about tumor cells, such as mutant status, apparent status, and expression levels of related proteins, may be expressed in a few or a single cell. Heterogeneity can be ignored if a mixture of tumor cells is used for analysis. Compared with previous sequencing methods, this method can more comprehensively analyze healthy cells and tumor cells, and identify previously unknown cell types, which can better reveal the heterogeneity of tumor cells at the cellular and molecular levels. Therefore, scRNA-seq plays a significant role in studying the individual differences in cells involved in the occurrence of GC.

Apolipoprotein E (APOE) is a multifunctional protein involved in lipid transport and lipoprotein metabolism that mediates the distribution/redistribution of lipids in tissues and cells [8]. APOE can also regulate inflammation and immunity, maintain the stability of the cytoskeleton and improve the function of neural tissue [9]. Studies have shown that APOE can regulate polarization in macrophages, participate in the construction of the tumor immune microenvironment, regulate tumor inflammation and the immune response and play a role in tumor progression, invasion, and metastasis [10–12]. However, the relevant regulatory mechanisms of APOE in GC are still not fully understood, therefore, further in-depth study is necessary.

2 Materials and methods

2.1 Data extraction and collation

The Cancer Genome Atlas (TCGA) project (<https://www.cancer.gov/tcga>) is an open database which aims to make molecular data characterizing the cancer-related genome freely available and link genomic data to patients' clinicopathological information. The expression profiles and relevant clinical information samples of stomach adenocarcinoma (STAD) were downloaded from TCGA databases. In addition, the Affymetrix microarray data sets GSE183904, GSE84437 and GSE26253 were obtained from Gene Expression Omnibus (GEO) database (<https://www.ncbi.nlm.nih.gov/geo/>). GSE26253 datasets were used for external validation. After background adjustment and quantitative normalization of all the downloaded data, the TCGA-STAD dataset was combined with the GSE84437 dataset (N = 804), Table 1 showed the details of expression datasets involved in the study. The "Combat" function was used to reduce the variance between different datasets in the possibility of batch effects caused by non-biological factors and technical deviations. The clinical variables included in this study were age, sex, T stage, N stage, survival status and survival time. In addition, a scRNA-seq count matrix was obtained from the GSE183904 dataset. This dataset covered 40 samples, including 29 primary tumor samples and 11 corresponding paracancerous samples.

2.2 Single-cell quality control and clustering

The GSE183904 dataset was selected from the GEO database and subjected to quality control (QC). It is finished by the "Seurat" package. We used "PercentageFeatureSet" function to calculate the percentage of mitochondrial genes, and the

Table 1 The patient characteristics of STAD and GSE84437 datasets

Group	Tumor (N = 772)	Normal (N = 32)	P
Event			0.211
Dead	281 (36.4%)	4 (12.5%)	
Alive	491 (63.6%)	28 (77.5%)	
Age			0.007
≤ 65	439 (56.8%)	11 (34.3%)	
> 65	333 (43.2%)	21 (65.7%)	
Gender			0.092
Male	254 (32.9%)	10 (31.2%)	
Female	518 (67.1%)	22 (68.8%)	

top 2000 genes with highly variable features are identified by ANOVA. In the case of false discovery rate (FDR) < 0.05, Principal component analysis (PCA) was used to filter the significantly separated dimensions, and then the t-SNE algorithm was used to downscale the first 15 principal components (PCs) to obtain the main cluster. Specifically, PC was determined by the “JackStraw” function. Then the marker genes in each cluster were screened for log2 FC > 1 and FDR < 0.05, and the first 10% of the marker genes in the first 15 clusters were spread on the heat map.

After initial dimensionality reduction and clustering on the PCs, 36 subclusters were obtained. The “SingleR” package and manual annotation were used to match different cell clusters. Each cluster was defined based on comparative analysis and the cellular markers identified. A UMAP plot and a t-SNE map were drawn to evaluate cell clustering.

2.3 Secondary dimensionality reduction and clustering

The “CreateSeuratObject” function of “Seurat” was used to cluster immune cells and myeloid cell subsets again, and 20 cell clusters were obtained. After classifying the expression of the top 50 markers in different clusters, the cell subsets of the secondary clusters were labeled. The “ggrepel” package was used to evaluate the proportions of different cell subsets in GC and normal samples. Next, the expression of immunosuppressive genes in cell subsets was plotted using the “DotPlot” function in the “ggplot2” package.

2.4 Cell trajectory inference

Single-cell trajectory inference was performed using Monocle2. The gene selected criteria were as follows: expression in more than five cell subsets and the average expression level was > 0.3, the P value was < 0.05. DDRTree is a manifold learning algorithm designed to embed primary graphs into high-dimensional single-cell RNA-seq data to find high-dimensional gene expression space mapping to low-dimensional space. We build a minimum spanning tree (MST) with DDRTree by setting the number of reduced dimensions to 2, project the cells onto the MST and calculate the pseudotime trajectory. The cell trajectories between CD8+ T cells and monocyte/macrophage subsets were visualized using the “Plot cell trajectory” function.

2.5 Cell interaction analysis

To better clarify the associations of ligands and receptors with different cell subsets, the interaction strength and receptor–ligand interaction relationship of cell subsets were visualized using the “netVisual” function in the “CellChat” package to assess the interaction of each cell subset in tumor and normal samples.

2.6 Single-gene pan-cancer analysis

First, the STAD and GSE84437 datasets were combined. The 3627 genes obtained from single-cell RNA sequencing and the DEGs obtained from the STAD-84437 dataset were intersected, yielding 945 common immune-related DEGs.

The high expressional genes of the monocyte/macrophage subsets were differentially screened using the “FindAllMarkers” function with |LogFC| ≥ 1 and P < 0.05 as the screening criteria, yielding 62 genes. The genes and the immune-related DEGs intersected with, i.e., APOE.

In order to explore the expression of APOE in different cancers, pan-cancer analysis of APOE gene was performed using “ggpubr” package of R software with $|\text{LogFC}| \geq 1$ and $P < 0.05$ to evaluate the expression of APOE among different tumor samples. Then, the frequency and types of APOE mutations in different cancers were investigated by using the cbiportal database (<https://www.cbiportal.org/>). Next, we obtained the immune and molecular subtypes of APOE in gastric cancer in the TISIDB database (<https://cis.hku.hk/TISIDB/>). We analyzed the relationship between APOE, immunity, and molecular subtypes by using preprocessed gene expression data from the TISIDB database and phenotypic information of TCGA cancers.

2.7 Forest plot of hazard ratios

By constructing tuning parameter and coefficient diagrams, we developed an APOE-related prognostic survival model and evaluate the reliability by using the “ggforest” function in the “survminer” package. We get a prognostic survival model constructed using six genes, including APOE, the correlation between the model score and clinical factors was evaluated. A risk factor map was constructed to analyze the effect of the risk score on the survival of patients and the correlation of prognosis with the risk score was analysed in STAD, GSE84437 and combined dataset.

2.8 Immune infiltration analysis and enrichment analysis

Immune infiltration analysis and functional enrichment analysis were performed on the DEGs in the high-risk and the low-risk group by using “cibersort” and “enrichplot” packages to assess the differential immune infiltration and enrichment between the different groups.

2.9 Cell culture and transfect with siRNA

We obtained the gastric cancer cell lines of HGC-27 and SGC-7901 from Chinese Academy of Sciences (Shanghai, China). The cell lines were cultured in RPMI-1640 medium (Gibco, USA) with 10% FBS (fetal bovine serum) (Gibco, USA) in a cell incubator (37 °C, 5% CO₂). These cell lines were authenticated by short tandem repeat analysis to make sure the accuracy and reliability. APOE small interfering RNA (siRNA) and control siRNA (Thermo Fisher, USA) were transfected into GC cells by the Lipofectamine™ 2000 (Thermo Fisher, USA). The target sequence for APOE siRNA was 5'-CAUGGAGGACGUGUGCGGCCGCCUG-3'. The target sequence for control siRNA was 5'-CGGCCGCCAUGUUGCAACCGGGAAG-3'.

2.10 CCK-8 assay and drug sensitivity analysis

Cells were plated in 96-well microtiter culture plates at a density of 5×10^3 cells per well and incubated at 37 °C overnight. After APOE expression was inhibited by siRNA for 12 h, 24 h and 36 h, CCK-8 was added to each well. Cells were incubated for another 4 h and scanned at 470 nm with a multiwavelength spectrophotometer (Thermo Fisher, USA). To identify potential drugs that target APOE, the CellMiner database (<https://discover.nci.nih.gov/cellminer/home.do>) was used to analyze the relationship between APOE expression and drug sensitivity.

2.11 Scratch test

After the cells were plated in 6-well plate overnight, each well was scratched mechanically with sterile pins. After taking pictures of the scratches at 0 h underneath an inverted microscope, cells were washed twice with phosphate buffered saline (PBS), the cell culture medium was replaced with basic RPMI-1640 medium and put back into the incubator. After 24 and 36 h, the scratches were photographed again. Image J and GraphPad Prism 10 (GraphPad Software, Inc.) were used to calculate wound areas and for statistical data analysis, respectively.

2.12 Apoptosis analysis

Apoptosis analysis of GC cells was assessed using an Annexin V-fluorescein isothiocyanate (FITC)/propidium iodide (PI) apoptosis detection kit (Beyotime, China). Briefly, cells were transfected APOE siRNA for 24 h. Stained cells were analyzed within 1 h using a FACScan® flow cytometer (BD Biosciences, USA).

3 Results

3.1 Evaluation of tumor microenvironment by Single-cell RNA sequencing

In this experiment, the GSE183904 dataset in the GEO database, which included 29 tumor samples and 11 normal samples, containing a total of 158,641 cells, was analyzed. After QC of 40 samples, cells with < 500 genes, < 60 sequencing reads and mitochondrial gene content > 5% were excluded to reduce data errors (Figure S1). After PCA was used to filter the dimensions with significant separation, a total of 36 PCs were obtained. After that, the t-SNE algorithm was used to perform dimensionality reduction analysis on the former PCs and the top 10% HVGs in the top 15 clusters were presented in heatmaps (Fig. 1A).

Next, initial dimensionality reduction and clustering were performed for PCs by using the SingR package and manual annotation [13]. Subsets of 10 different cell types were obtained, including T cells, B cells, NK cells, myeloid cells, epithelial cells, endothelial cells, fibroblasts, tissue stem cells, smooth muscle cells and liver cells (Fig. 1B, C). Then, secondary dimensionality reduction and clustering were performed for immune cell and myeloid cell subsets. The secondary clustering results showed that the immune cell subsets were predominantly immune regulatory cells, such as naive T cells, CD8⁺ T cells, regulatory T cells, and monocytes/macrophages (Fig. 2A, B), the marker genes of each cell subset were

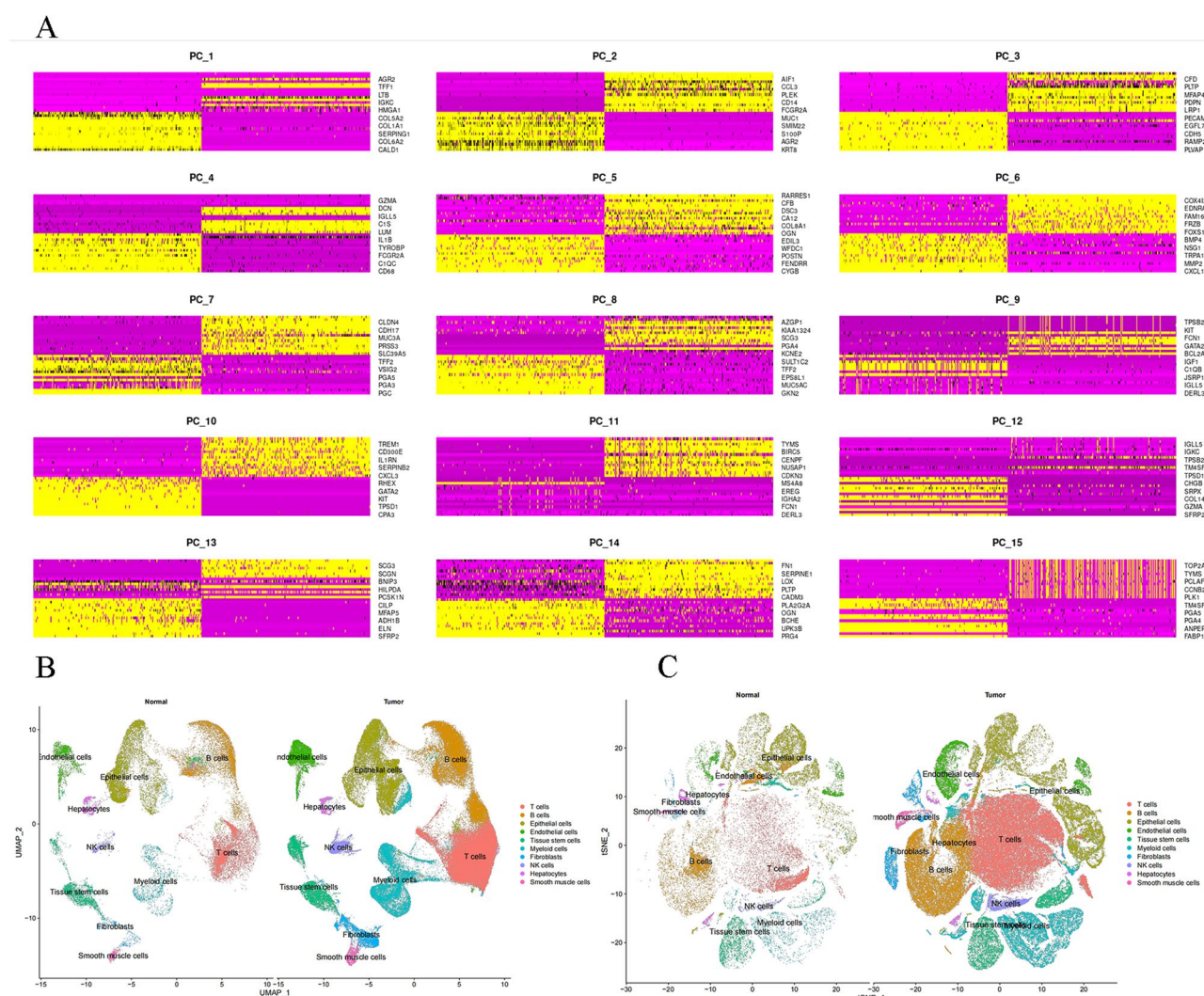


Fig. 1 Single-cell quality control and clustering. **A** Correlation heatmaps of the top 15 clusters. **B** UMAP plot after the initial dimensionality reduction and clustering. **C** t-SNE diagram after initial dimensionality reduction and clustering

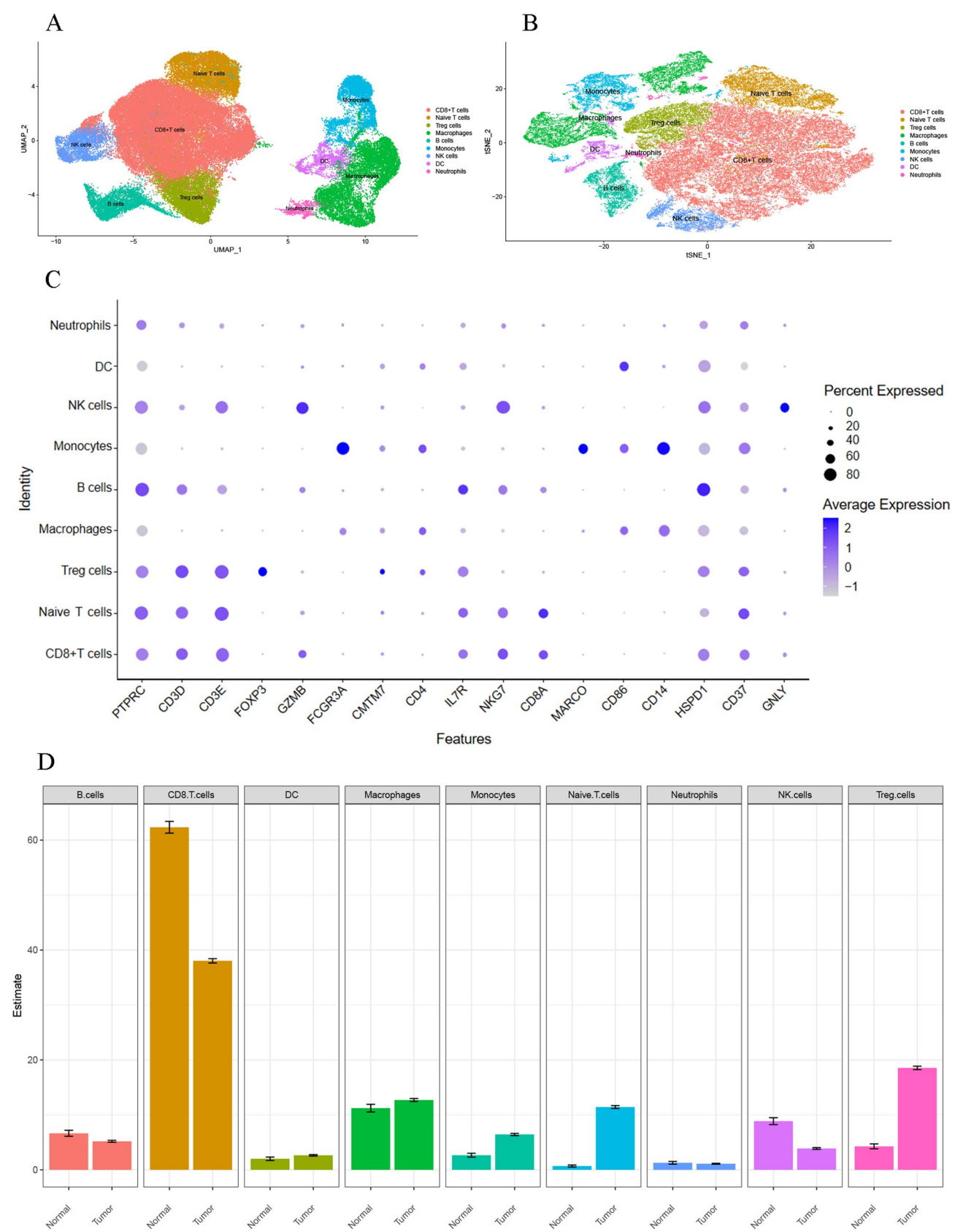


Fig. 2 Secondary dimensionality reduction and clustering. UMAP (**A**) and t-SNE (**B**) plots, nine clusters were annotated based on marker genes. **C** Marker gene expression in different cell subsets. **D** Analysis of the proportions of different cell subsets in normal tissue and GC tissue

shown as Fig. 2C. The intergroup comparison showed that compared with normal samples, samples from GC patients had significantly fewer CD8+ T cells and more regulatory T cells (Fig. 2D), indicating that the tumor microenvironment (TME) of GC may be associated with immune exhaustion [14, 15]. Therefore, immune checkpoint inhibitor genes in the cell subsets obtained by clustering were analyzed (Figure S2). The results showed that compared with the normal cell subsets, the expression of immunosuppressive subset-associated genes such as PDCD1, LAG3, HSPD1 and CTLA4 in the GC cell subsets was significantly higher, indicating that the tumor microenvironment of the cell subsets obtained from secondary clustering tended to be immunosuppressive and cancer cells may promote the development of GC through immune evasion [16].

3.2 Pseudotime analysis of CD8+ T cell and monocyte/macrophage subsets

Since CD8 T cells and mononuclear/macrophage subsets accounted for the largest proportion in the subsets of secondary clustering and were the cell subsets that played a major role in immunosuppression. We selected them for further study in order to evaluate different cells in the process of growth differentiation (Fig. 3A, C). As the results showed in Fig. 3B, PDCD1, LAG3 and other immunosuppressive genes in CD8+ T cell subsets were significantly expressed in each cluster, indicating that the differentiation of CD8+ T cell subsets as a whole tended towards immune exhaustion. Apart from that, genes about natural killer immune cells such as CD8A and CD8B were significantly expressed in Cluster 1, Cluster 3, and Cluster 6. These data suggest that the CD8+ T cell subsets mainly exist in the development-killing-exhaustion mode [17]. In contrast, pseudotime analysis and trajectory inference of monocyte/macrophage subsets was showed in Fig. 3D. Marker gene expression was dominated by macrophage immunosuppressive genes such as CD206 and MRC1 [18].

3.3 Cell interaction analysis of monocyte/macrophage subsets

Next, cell interaction analysis was performed to further investigate the relationships between different cell subsets in GC and normal samples. The results indicated that there were 1241 cell interaction pairs in the GC samples, with slightly fewer cell interactions in the normal samples (1147 pairs). The monocyte/macrophage subsets had a high number of interactions with other cell subsets (Fig. 4A), indicating that they had good interactions with other cell subsets. Therefore,

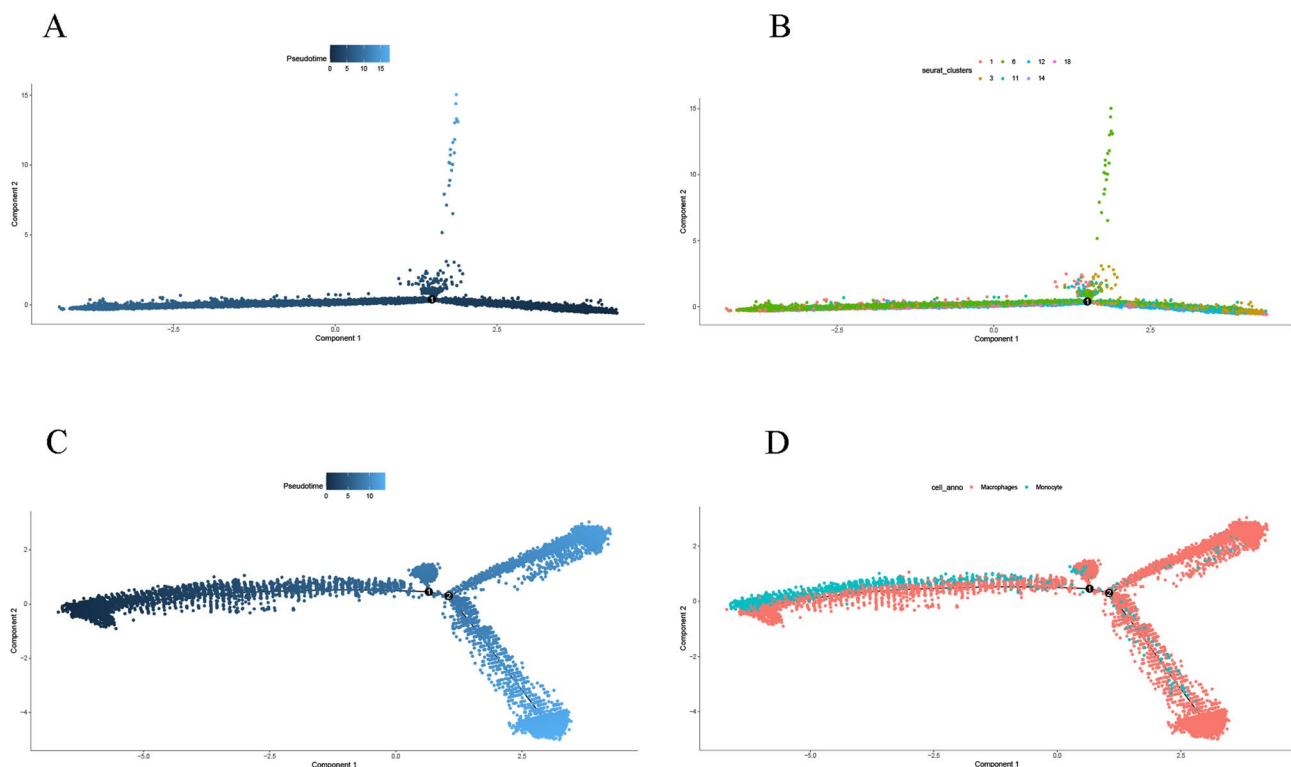
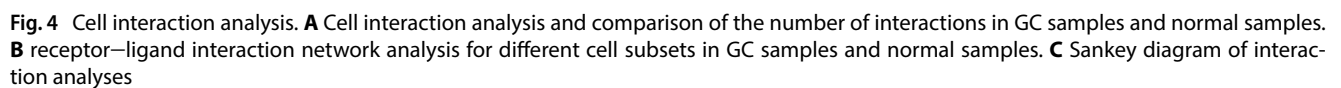


Fig. 3 Cell trajectory inference. **A, B** Pseudotime analysis of CD8+ T cells. **C, D** Pseudotime analysis of monocytes/macrophages



we focused on the differential expression of other immunosuppressive cell subsets and monocyte/macrophage subsets between GC and normal samples (Fig. 4B). The results showed that compared with normal tissues, GC tissues had significantly more receptor–ligand pairs, such as ADGRE5-CD55, APP-CD74 and ALCAM-CD6, indicating that these receptor–ligand pairs may have potential effects on the occurrence and development of GC. Streamgraph analyses showed that the transcription factors were mainly expressed in monocytes/macrophages and CD8⁺ T cells (Fig. 4C).

3.4 APOE was identified as a hub gene and correlated with M2 macrophages

The STAD and GSE84437 datasets were downloaded from the TCGA database and the GEO database respectively. After merging the two datasets and removing the batch effect, there were 2148 upregulated genes and 1454 downregulated genes. The DEGs obtained from STAD-84437 and the 3627 DEGs obtained from single-cell RNA sequencing were intersected, yielding 945 common DEGs (Figure S3).

Cell interaction analysis showed that macrophages had a strong association with other immunosuppressive cell subsets. Therefore, HVGs of macrophage subsets were separately extracted to analyze their differential expression in macrophage subsets by using $|\text{Log}_2\text{FC}| > 1$ and $P < 0.05$ as the screening criteria. The results showed that there were 62 DEGs when comparing the macrophage subsets in normal and GC tissues. The HVGs were associated with macrophage subsets and immune-related DEGs were intersected, yielding one upregulated gene, APOE. Because previous experimental results showed that most macrophage subsets were significantly enriched in immunosuppressive macrophages, M2 macrophage marker genes and APOE were selected for gene association analysis. The results showed that APOE was significantly positively correlated with M2 macrophage marker genes such as CD206 (MRC1), CCL17, IL10, CD163 and CLEC7A [19–22] (Figure S4), indicating that APOE expression was closely related to macrophages and may promote the occurrence and progression of GC by promoting M2 macrophage gene expression.

3.5 Pancancer analysis of APOE

Considering that APOE may also play an important role in other cancers, pancancer analysis was performed using the TCGA database. The results showed that APOE was widely expressed in various gastrointestinal tumors, such as GC, liver cancer, colon cancer, and esophageal cancer (Fig. 5A). Compared with normal tissues, the expression of the APOE gene was significantly higher in GC and esophageal cancer but lower in CRC, lung cancer and pancreatic cancer (Figure S5). Next, the mutation frequency and corresponding mutation types of APOE in different cancers were further analyzed (Fig. 5C) and the mutation characteristics and location of APOE in tumors were investigated (Fig. 5B). The results showed that APOE mutations, mainly nonsense mutation, were most prominent in GC and cutaneous melanoma. Furthermore, APOE was significantly upregulated expression in GC (Fig. 5D) and associated with GC stage (Fig. 5E), mainly manifesting as relatively low APOE expression in early GC and significantly increased in advanced GC. In terms of immune subtypes, APOE mainly expressed IFN- γ in GC (Fig. 5F), high APOE expression could change the tumor microenvironment through the expression of IFN- γ and promote the transformation of macrophages from the pro-inflammatory M1 phenotype to the anti-inflammatory M2 phenotype [23]. APOE was not significantly differentially expressed among GC molecular subtypes (Fig. 5G). Therefore, the expression level of APOE is correlated with the clinicopathological stage and immune subtype of tumors. The results of the above experiments indicated that APOE may play a broad regulatory role in different cancers by regulating the development of macrophage-induced immunosuppression.

3.6 Construction of APOE related survival model

Prognosis-related risk models of GC were screened by using tuning parameter and tuning coefficient diagrams, we constructed a prognostic survival model contained with six genes, i.e., APOE, SNRPN, RAB32, CMTM7, RASGEF1B and CLEC10A and evaluated its accuracy. ROC curves showed that the area under the curve (AUC) values for the prognostic risk model were 0.75 and 0.73 (Figure S6), which indicated that the model was accurate and effective. Next, a forest plot of hazard ratios was drawn to evaluate the correlation between the survival model and the clinical factors of patients. The results showed that the APOE-related survival model was closely related to the T and N stages, age, and sex of the patients (Fig. 6A). Due to the random error of the survival model, we introduced dataset GSE26253 as an external validation cohort, our results showed that the APOE survival model was significantly correlated with clinical stage and the prognosis of gastric cancer (Figure S7). Among GC patients grouped based on the risk score of the APOE-related survival model, the high-risk group showed a higher number of deaths and expression levels of APOE, SNRPN, and RAB32 genes

Fig. 5 The expression of APOE in different cancers. **A** APOE was high expression in different cancers. **B** Mutation locus of APOE. **C** APOE had high mutation frequency in different cancers. **D** APOE expression in gastric cancer. **E** Association between APOE expression and different stages of GC. **F** Relationship between APOE expression and immune subtypes of STAD. **G** Relationships between APOE expression and molecular subtypes of STAD

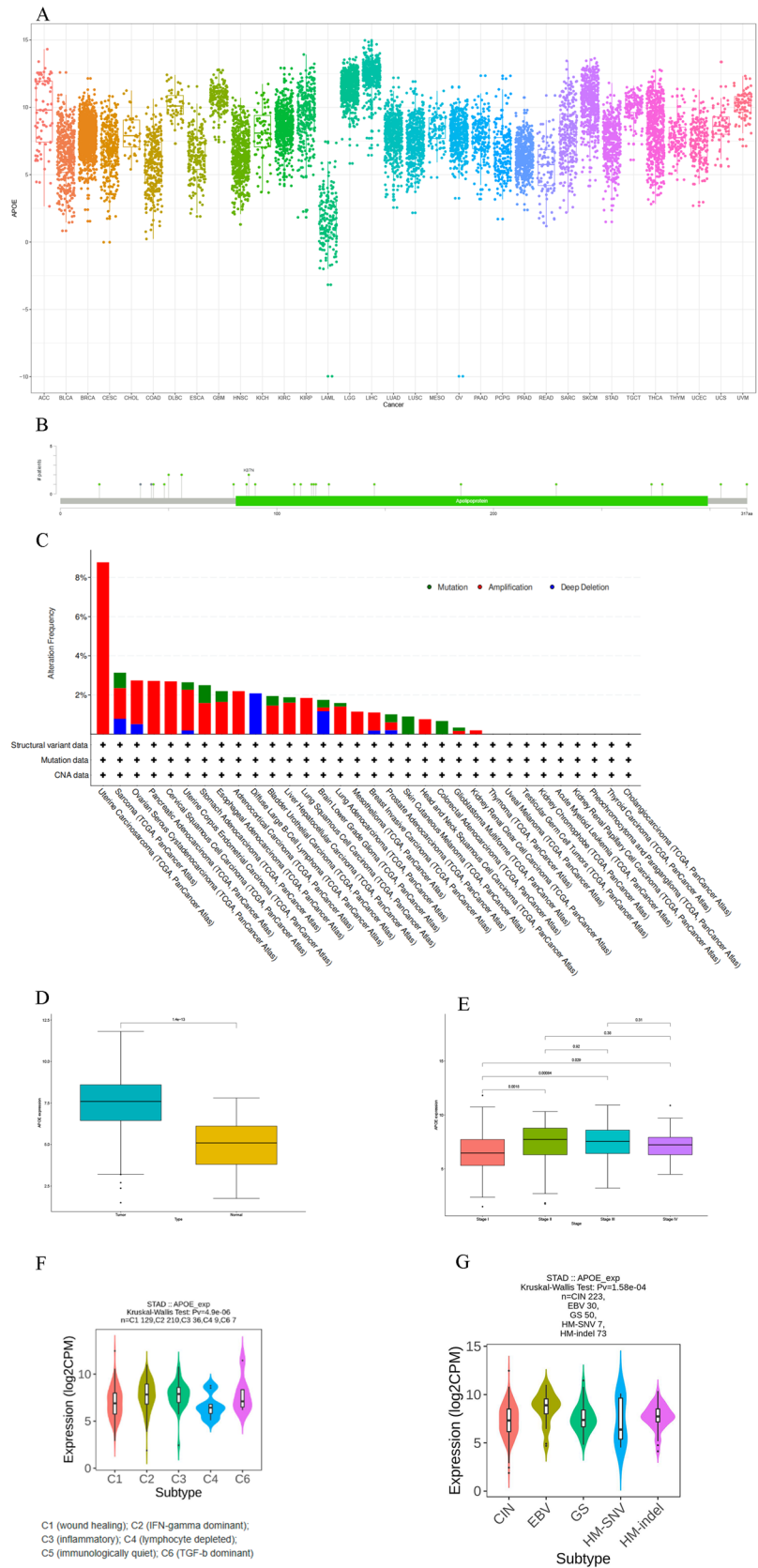
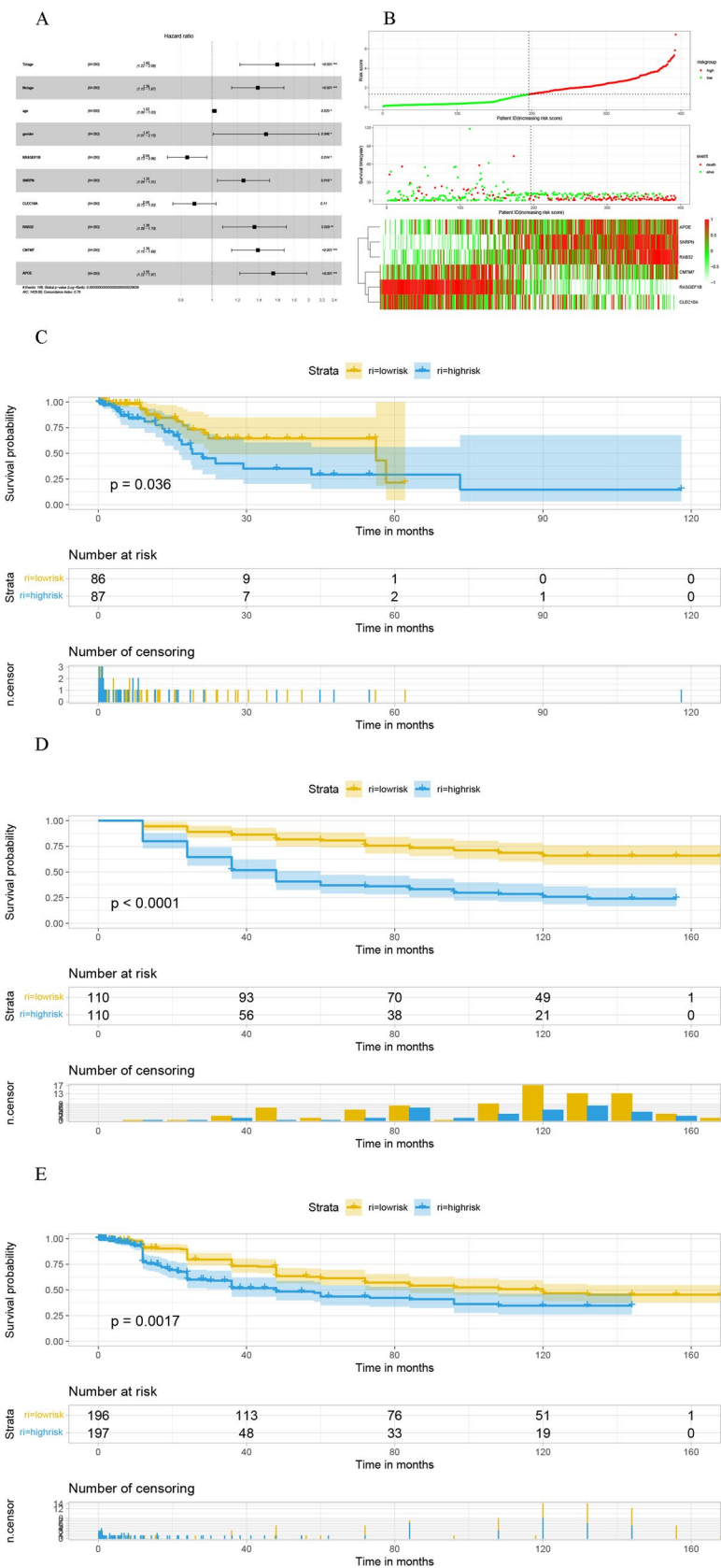


Fig. 6 Survival model of APOE. **A** Forest plot of hazard ratios of the GC survival model. **B** Triple plot of risk factors for the survival model in the high-risk and low-risk groups. Analysis of the correlation between GC survival model score and prognosis in the STAD (**C**), GSE84437 (**D**), and combined dataset (**E**)



than the low-risk group (Fig. 6B), indicating that these genes are adverse prognostic factors in GC patients. Based on the prognosis analysis of STAD, GSE84437 and combined dataset (Fig. 6C–E), the low-risk group had a better prognosis than the high-risk group, which was consistent with our previous experimental results.

3.7 Immune infiltration and enrichment analysis of APOE related model

To further assess the difference between the models for the high- and low-risk groups, immune infiltration analysis and enrichment analysis were performed. The immune infiltration analysis results showed that the differentially enriched immune cells between risk groups were mainly plasma cells, chief cells, and B cells (Figure S8). Compared with the low-risk group, the high-risk group had fewer CD8⁺ T cells but more M2 macrophages, the result was consistent with our previous experimental findings. The number of chief cells, naive B cells and dendritic cells was higher in the high-risk GC group (Fig. 7A). In contrast, the number of CD8⁺ T cells and M0 and M1 macrophages was higher in the low-risk GC group than in the normal group (Fig. 7B), indicating that the tumor microenvironment in the high-risk group might be associated with immunosuppressive.

The enrichment analysis results showed that the activation of myeloid cells, chemotaxis of tumor necrosis factors, macrophage differentiation and migration, signaling receptor activation, and cytokine activity played important roles in the high-risk group (Fig. 7C). In the low-risk group, the processes such as the activation of the immune response, lymphocyte differentiation, oxidative phosphorylation and cellular redox were significant (Fig. 7D). The enrichment results for the low-risk and low-risk groups further demonstrated that the model we constructed was associated with the immune response, possibly having certain guiding significance in the clinical application of immunotherapy for GC.

3.8 Sensitivity analysis of APOE and drugs

In the CellMiner database, APOE expression was negatively correlated with the sensitivity to most drugs (such as colchicine, aclarubicin, docetaxel, and palbociclib) (Fig. 8A–D). and positively correlated with cinchonidine, vemurafenib, abexinostat, and hypothemycin (Fig. 8E–H).

3.9 APOE promoted the growth of GC cells

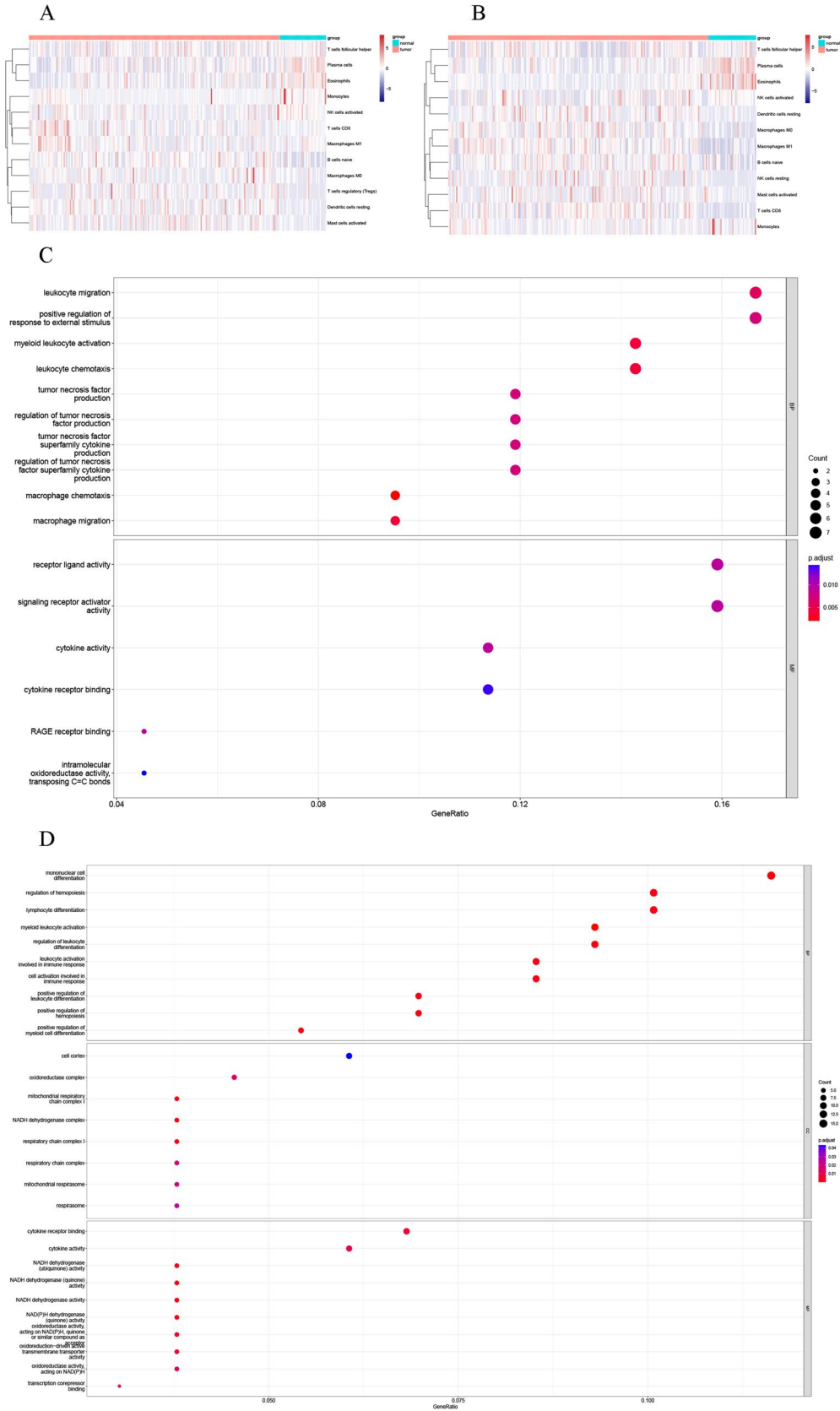
To explore the effect of APOE on growth of HGC-27 and SGC-7901 cells, we used si-APOE to transfect HGC-27 and SGC-7901 cells in order to inhibit the expression of APOE (Figure S9). As shown in Fig. 9A, the viability of GC cells was significantly inhibited after transfection with si-APOE than control group. Moreover, the scratch test (Fig. 9B) showed that migratory ability was significantly suppressed after inhibiting the expression of APOE. After transfection with si-APOE for 24 h, our results showed that the apoptosis of GC cells significantly increased after inhibiting the expression of APOE (Fig. 9C).

4 Discussion

GC is one of the most common malignancies worldwide. The prognosis of GC is poor due to the lack of specificity and sensitivity in diagnosis. In the past decade, immune checkpoint blockade has become an attractive treatment strategy for GC. Previous studies have shown that the tumor immune microenvironment of cancers exhibits genetic and transcriptional diversity and plays an important role in tumor progression, metastasis, and drug resistance. Therefore, exploring the immune microenvironment of GC from multiple angles and mining new immune-related genetic models will help in further understanding the immune pattern of GC and predicting the immunotherapy response of patients. In addition, single-cell RNA sequencing technology is developing rapidly and can be used to finely describe the heterogeneity within tumors, combining single-cell RNA sequencing and transcriptome data is a novel and reliable method to analyze the immune microenvironment status of GC.

In this study, secondary dimensionality reduction and clustering based on single-cell RNA sequencing results were performed to obtain clusters of cell subsets dominated by immune cells. To further clarify the composition of the immune microenvironment in terms of single-cell subsets, checkpoint gene detection was performed. The results showed that the CD8⁺ T cell immunosuppression-related gene LAG3, the macrophage marker gene CTLA4 and the regulatory T cell related gene FOXP3 were universally expressed in their corresponding cell subsets [24, 25]. In addition to that, immunosuppressive genes, predominantly PDCD1 and HSPD1, were ubiquitously expressed in all immune cell subsets, indicating

Fig. 7 Immune infiltration and pathway enrichment of APOE in high and low risk group. **A** Analysis of the levels of infiltrating immune cells in the high-risk group. **B** Analysis of the levels of infiltrating immune cells in low-risk group. **C** GO enrichment of the differential expressed genes between the the high (C) and low-risk (D) groups based on the survival model



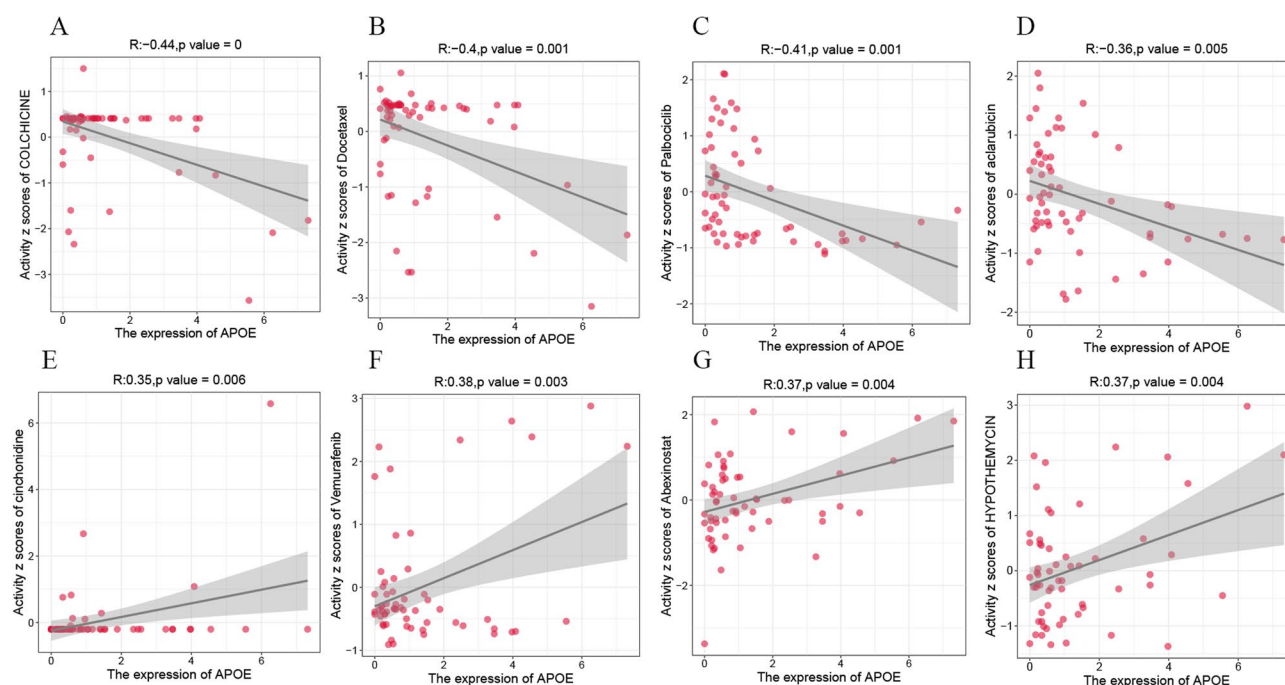


Fig. 8 APOE was sensitive to colchicine (A), docetaxel (B), palbociclib (C), aclarubicin (D), cinchonidine (E), vemurafenib (F), abexinostat (G) and hypothemycin (H)

that the tumor microenvironment might be inclined to immune exhaustion. During the development of GC, tumor cells may promote the development of immune exhaustion by excessively activating immunity, which in turn triggers tumor escape to promote GC growth.

The cell trajectory inference was performed for different T cell subsets and monocyte/macrophage cell subsets to clarify the growth and differentiation modes of different cell subsets. The results showed that in the CD8⁺ T cell subsets, the immunosuppressive genes, predominantly PDCD1 and LAG3, were ubiquitously expressed during the cell differentiation process, indicating that the CD8⁺ T cell subsets as a whole tended towards immune exhaustion and the genes about natural killer immune cells such as CD8A and CD8B were mainly expressed in Cluster 6, indicating that they might be inclined towards a killing mode. Then, the mode of differentiation of CD8⁺ T cells might manifest as the development-killing-exhaustion process [26]. CD8⁺ T cells develop excessively due to stimulation by tumor growth in the early stage of GC, thus killing tumor cells through an immune mechanism. In the late stage of tumorigenesis, due to the long-term excessive activation of the immune response, immune exhaustion occurs and the generated immune cells cannot effectively and completely eliminate tumor cells, which in turn promotes the growth of tumor cells [27].

Next, to clarify the interaction between different immunosuppressive cell subsets, cell interaction analysis was performed. The results showed that the number of pairings between the monocyte/macrophage subsets and other cell subsets was high, it suggested a good interaction relationship between them. Therefore, we focused on the pairing relationship between monocyte/macrophage subsets and other immunosuppressive cell subsets. As seen in the ligand–receptor-mediated cell interaction diagram, compared with normal tissues, GC tissues had significantly higher ADGRE5-CD55, APP-CD74 and ALCAM-CD6 receptor–ligand pairings in monocyte/macrophage subsets and other paired subsets. Notably, ADGRE5 and APP are considered associated with inflammation and ALCAM, also called leukocyte adhesion factor, binds to CD6 to regulate the immune balance by causing the activation and proliferation of immune cells. However, the regulatory mechanism of the CD6-ALCAM pathway in GC remains unclear. The overall correlation analysis between GC and normal tissue showed that the expression of the collagen and CSF genes in GC tissue was significantly higher than that in normal tissue and the expression of MHC-I in GC tissues was lower than that in normal tissue. As seen in the stream graph, CSF was abnormally expressed in monocytes/macrophages, indicating that CSF may be a potential GC-affecting gene that regulates the occurrence and progression of GC mainly through monocytes/macrophages. These results suggest that mononuclear/macrophage interaction in GC tissues is associated with tumor inflammation and immune response. This lays a theoretical foundation for the conclusion that APOE gene may play an immunosuppressive role by promoting the expression of M2 macrophage gene, thus promoting the occurrence and progression of GC.

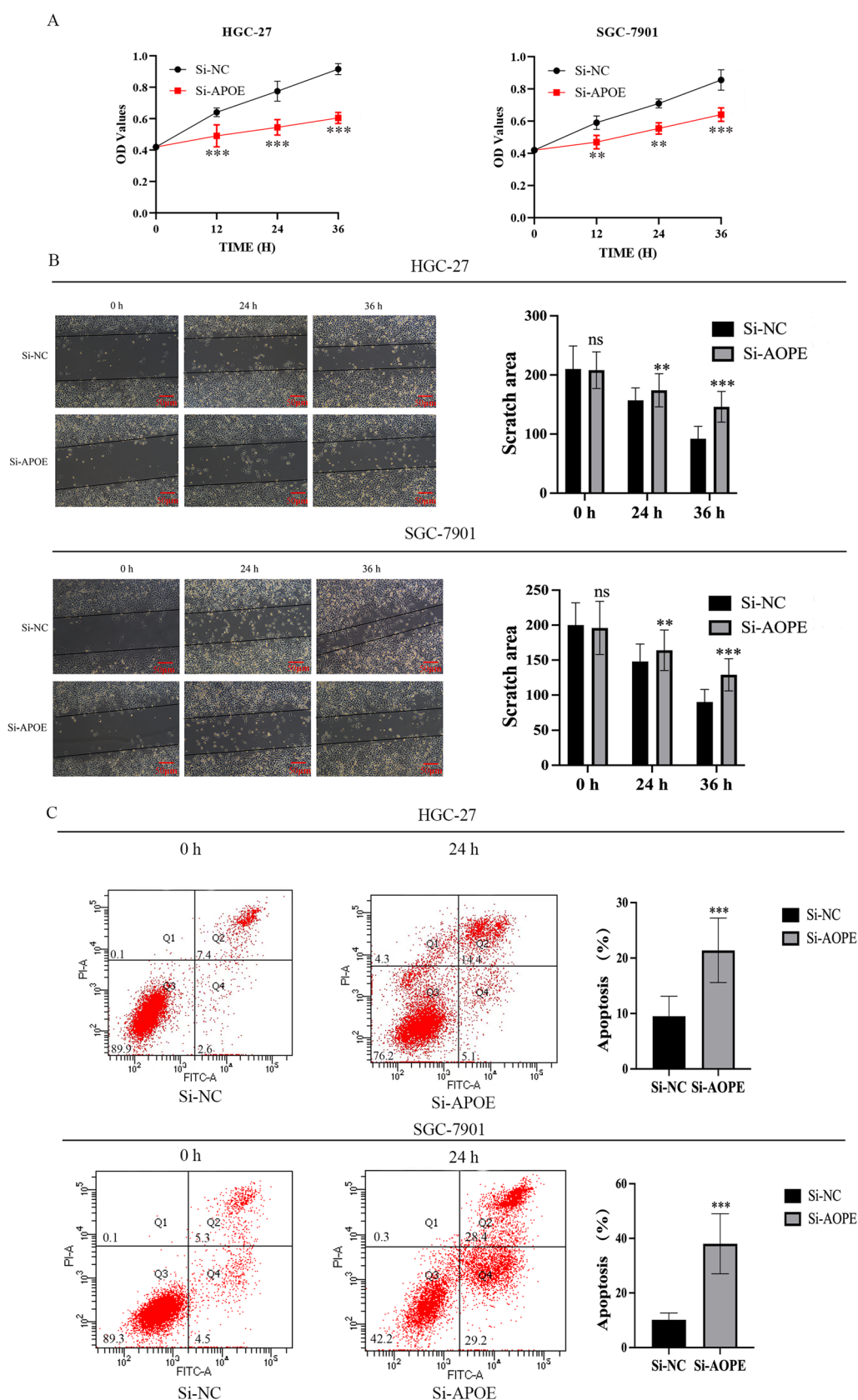


Fig. 9 APOE promotes proliferation of GC cells. **A** Cell viability was measured by CCK-8 assay. **B** Scratch test was used to assess migrated of GC cells after transfected with Si-APOE. **C** GC Cells apoptosis was analyzed by Annexin V-FITC/PI double staining in Si-NC and Si-APOE. All experiments were performed in triplicate. (** < 0.01; *** < 0.001)

Because the levels of macrophage subsets and other cell subsets were highly correlated, the macrophage subsets were selected for further analysis. Based on the differential expression of immune-related HVGs in the macrophage subset between GC and normal samples, the hub gene APOE was identified. APOE is highly expressed in many cancers, including glioma, ovarian cancer and oral squamous cell carcinoma, it is significantly upregulated in the tumor extracellular matrix (TEM) [28, 29]. Tumor progression is closely related to chronic sterile inflammation and TEM instability [30]. The expression of APOE in the gastric TME is closely related to tumor-associated macrophages (TAMs). TAMs are the only APOE-synthesizing immune cells in the GC microenvironment [31]. M2 TAMs promote tumor proliferation and are associated with a poor patient prognosis for many cancers [32]. Previous data [33] showed that M2 macrophages promoted the migration and invasion of GC cells through epithelial-to-mesenchymal transition (EMT). Therefore, M2 macrophage marker genes associated with immunosuppression and APOE were selected for gene association analysis. The results showed that APOE was significantly positively correlated with M2 macrophage marker genes and that the expression of APOE in advanced GC was significantly upregulated compared with that in early GC. This finding indicates that the APOE gene may play an immunosuppressive role by promoting the expression of M2 macrophage genes, thus promoting the development and progression of GC.

Next, pancancer analysis was performed to investigate the expression of APOE in different cancers. The expression of the APOE gene was significantly higher in cancer tissues, such as GC, esophageal cancer, prostate cancer and thyroid cancer. The expression of the APOE gene was downregulated in colon cancer, rectal cancer, lung squamous cell carcinoma and pancreatic cancer. The results indicated that APOE may induced immunosuppression through the regulation of macrophages and play an extensive regulatory role in different cancers [34–36]. It may occur because more energy is needed to maintain tumor growth and because tumor cells obtain the energy needed for growth through abnormal lipid metabolism and aerobic glycolysis. APOE, as an important mediator in the regulation of cholesterol metabolism, participates in the regulation of lipid metabolism in the progression of different cancers [37], thereby promoting cancer progression. In this study, the expression of APOE was associated with the stage of GC. The results showed that the expression level of APOE was significantly increased in advanced GC. Based on an analysis of the molecular subtypes and immune subtypes of APOE in GC, we speculate that APOE may change the tumor microenvironment [38] through the expression of IFN- γ , which promotes the transition of macrophages from the pro-inflammatory M1 phenotype to the anti-inflammatory M2 phenotype, thus promoting GC growth. In advanced GC, due to the growth of GC cells, the energy needed further increases and the expression levels also increase [39].

A total of 945 common DEGs were obtained after intersecting STAD-84437 and GSE183904. Tuning parameters were screened to obtain a survival model composed of six genes: APOE, RASGEF1B, CLEC10A, CMTM7, SNRPN and RAB32. The survival model was assessed for its association with clinical factors, accuracy and prognostic efficacy. The results showed that the survival model was significantly associated with clinical factors such as T stage, N stage and age, the constructed model had high accuracy and a significant impact on prognosis. The model was used to validate the prognosis of GC patients in high-risk and low-risk groups in STAD, GSE84437 and combined dataset. The results showed that the prognosis of patients in the high-risk group was poor, while the patients in the low-risk group often had better outcomes. It indicated that the survival model can play an important role in the diagnosis and prognosis evaluation of GC.

Next, immune infiltration and enrichment analysis were performed for the survival model-associated genes in the high-risk and low-risk groups. As the result showed, compared with the low-risk group, the high-risk had fewer CD8⁺ T cells but more M2 macrophages, which is consistent with our previous experimental finding. In the high-risk group, the number of chief cells, naive B cells and dendritic cells was higher than low-risk group. In contrast, the number of CD8⁺ T cells and M0 and M1 macrophages in the low-risk group was higher, indicating that the tumor microenvironment might be dominated by immunosuppression [40], with the environment being predominantly immunostimulating in the low-risk group [41]. In addition, enrichment analysis suggested that the activation of myeloid cells, chemotaxis of tumor necrosis factors, macrophage differentiation and migration, signaling receptor activation and cytokine activity played important roles in the high-risk group, it suggested that the development of GC may be promoted through macrophage differentiation and receptor activation [42]. In the low-risk group, the activation of the immune response, lymphocyte differentiation, oxidative phosphorylation and cellular redox played important roles. The enrichment analysis results further demonstrated that the model we constructed was associated with the immune response, it may have certain guiding significance for clinical immunotherapy for GC.

The CellMiner database was used to better understand the correlation of drug sensitivity with APOE expression. The results showed that as the expression of APOE increased, the sensitivity to most chemotherapeutic drugs and targeted drugs decreased, some drugs liked cinchonidine and vemurafenib exerted better effects. Therefore, these findings could guide the clinical treatment of GC.

To further explore the potential role of APOE in GC, cell experiments were conducted to verify the function of APOE in GC. The results showed that after the inhibition of APOE expression by using siRNA, the growth rate of GC cells was significantly lower than that of cells in the control group, the level of migration ability was inhibited, the number of apoptotic GC cells increased. These findings indicate that APOE promotes the growth of GC cells and is an oncogene. The main cancer-promoting mechanism may be the induction of M2 macrophage activation. Unfortunately, due to the influence of various factors, we did not further study the relevant regulatory mechanisms in depth.

Finally, there are some biases and limitations in this study. Firstly, there is a bias between data sources and sample selection. (a): Diversity of data sources: model construction relies on specific datasets that may not represent the characteristics and backgrounds of all gastric cancer patients. As a result, the universality of the model may be limited. (b): Sample selection bias: there may be sample selection bias during the data collection process. Secondly, the limitations of bioinformatics analysis methods. (a): Limitations of algorithm selection: when constructing the model, we employed specific algorithms. However, different algorithms may have different advantages and disadvantages, and improper selection may limit the predictability of the model. (b): Quality and standardization of data: bioinformatics analysis involves a large amount of data processing and interpretation, but the quality and standardization of data have not been fully unified, which may affect the accuracy and reproducibility of the results. Thirdly, there are limitations in clinical application. Individual differences: although the model has demonstrated effectiveness and accuracy overall. But personally, due to physiological and pathological differences between patients, the predicted results of the model may not be completely consistent with reality.

5 Conclusion

In this study, we summarized the expression patterns of immune-related genes and explored their effects on the immune microenvironment, clinical features and prognosis of gastric cancer. We found that high expression of APOE and its survival model was significantly associated with poor prognosis of GC. In addition, APOE was closely associated with tumor immune microenvironment, it may induce M2 macrophage activation to promote the growth of GC. Therefore, targeting APOE may be an effective method for the treatment of gastric cancer. Further investigations are required to investigate the specific role by which APOE promotes the progression of gastric cancer.

Author contributions Zhi Zhao and Ni Li designed the research study. Ni Li, Jinquan Lin and Chunyu Wang conducted the experiments and analyzed the data. Xiao Ran prepared the reagents. Zhi Zhao wrote the main manuscript. All authors reviewed the manuscript.

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Data availability All the data are available on request from the corresponding author.

Declarations

Competing interests The authors declare no competing interests.

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